Econometrics for Business Economics

Chapter 2 - Bivariate analysis

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Chapter 2 - Bivariate analysis

1. Hypothesis tests
2. Relationship between two quantitative variables
3. Relationship between two categorical variables
4. Relationship between a quantitative variable and a categorical variable

Bivariate analysis

Bivariate analysis explores the relationship between two variables, the methods used depends on the type of variables being studied:

	quantitative variable	categorical variable
quantitative variable	linear correlation r	ANOVA
categorical variable	ANOVA	contingency table χ^2

These three methods will be studied in the following 3 sections, after introducing hypothesis tests.

Plan

1. Hypothesis tests
2. Relationship between two quantitative variables
3. Relationship between two categorical variables
4. Relationship between a quantitative variable and a categorical variable

Definition (Hypothesis test)

Rule of decision between two hypotheses about the value of a population parameter in a probabilistic model.

We define a **null hypothesis** H_0 about the value of an unknown population parameter. This is the hypothesis that is tested. We define also an **alternative hypothesis** H_1, which is most of the time the opposite of whatever is stated in H_0.

We want to be able to decide if the null hypothesis is true or false based on a sample drawn from the population.

The test is the rule of decision that conducts to **reject or not reject** H_0.

We consider two kinds of errors that can be made in hypothesis testing:

- Type I error or *level of significance* of the test, denoted α
- Type II error, denoted β.

reality \ decision	do not reject H_0	reject H_0
H_0 is true	correct conclusion	α Type I error
H_1 is true	β Type II error	correct conclusion

Type I error α is the probability of rejecting H_0 when it is true.

Only one error can be controlled. We specify α (generally $\alpha = 5\%$), the greater of the two errors, thus we control the risk of wrongly rejecting H_0.

1. Hypothesis tests (cont.)

We define a **test statistic**, which is a function of the sample of whom we know the probability distribution (Normal dist., Student's t-dist., Chi-square dist., Fisher F-dist.) under H_0.

The **p-value** (given by the statistical software) is the probability of getting a value equal to or more extreme than the sample's test statistic, assuming that H_0 is true.

To make a decision, we compare the p-value to α:

- if p-value $\leqslant \alpha$, we reject H_0,
- if p-value $> \alpha$, we do not reject H_0.

The test is qualified by **significant** if H_0 is rejected, and **not significant** (ns) otherwise.

For more information about hypothesis tests, see chapter 9 of the book *Anderson et al, Statistics for business and economics, Cengage, 5th edition, 2019.*

Data set *employees*

In this chapter, we will use the following variables of the data set *employees*:

- *gender*: Gender of the employee (1 pour *Female*, 2 pour *Male*)
- *pro_status*: Professional status (1 if *Worker*, 2 if *Security agent*, 3 if *Manager*)
- *salary*: Actual annual salary (in *k*$)
- *start_sal*: Annual starting salary in the company (in *k*$)

Population = employees of an American company, n=473.

Plan

1. Hypothesis tests

2. Relationship between two quantitative variables
2.1 Scatter plot
2.2 Covariance
2.3 Linear correlation coefficient
2.4 Pearson correlation test

3. Relationship between two categorical variables

4. Relationship between a quantitative variable and a categorical variable

2. Relationship between two quantitative variables: 1) Scatter plot

To study the relationship between two quantitative variables, we have to choose:

- a **dependent/response variable**, denoted Y
- an **independent/explanatory variable**, denoted X

Scatter plot

It is a graph that represents by a point each couple of observations (x_i, y_i) for $i = 1, \ldots, n$.

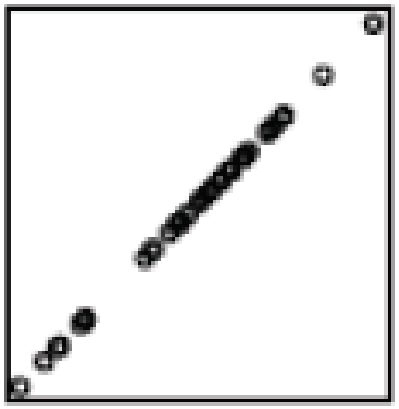
- Y-axis: variable Y
- X-axis: variable X

2.1 Scatter plot (cont.)

Pattern of points $\Rightarrow$ kind of relationship between X and Y:

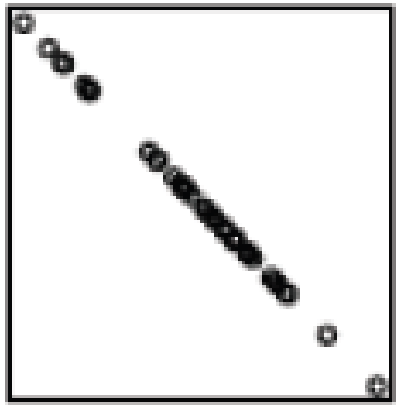
- **perfect linear correlation**: points fall on a straight line, Y is an affine function of X: $Y = aX + b$ (with a and b two constants)

scatter plot 1



positive correlation

scatter plot 2

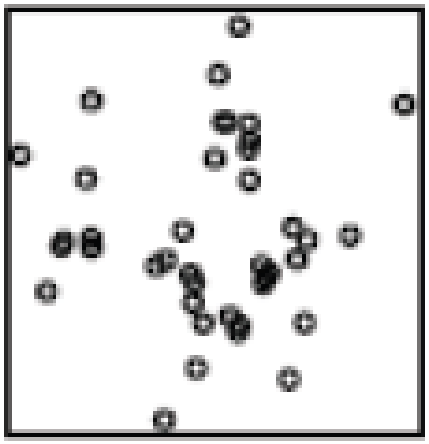


negative correlation

2.1 Scatter plot (cont.)

- **no correlation**: no pattern, no relationship between X and Y

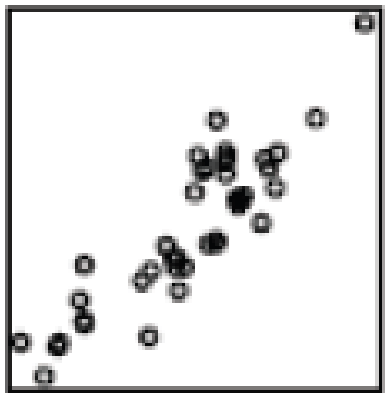
scatter plot 3



2.1 Scatter plot (cont.)

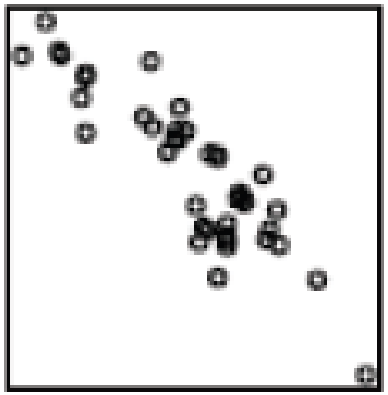
- **linear correlation**: points are distributed around a straight line

scatter plot 4



positive correlation

scatter plot 5

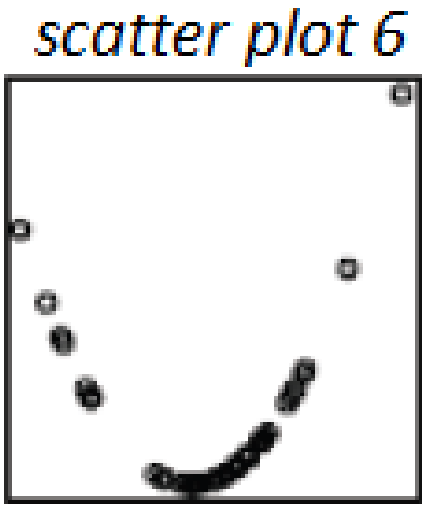


negative correlation

Note that stronger relationships produce a tighter clustering of points.

2.1 Scatter plot (cont.)

- **non linear relationship**: points are distributed around a curve line

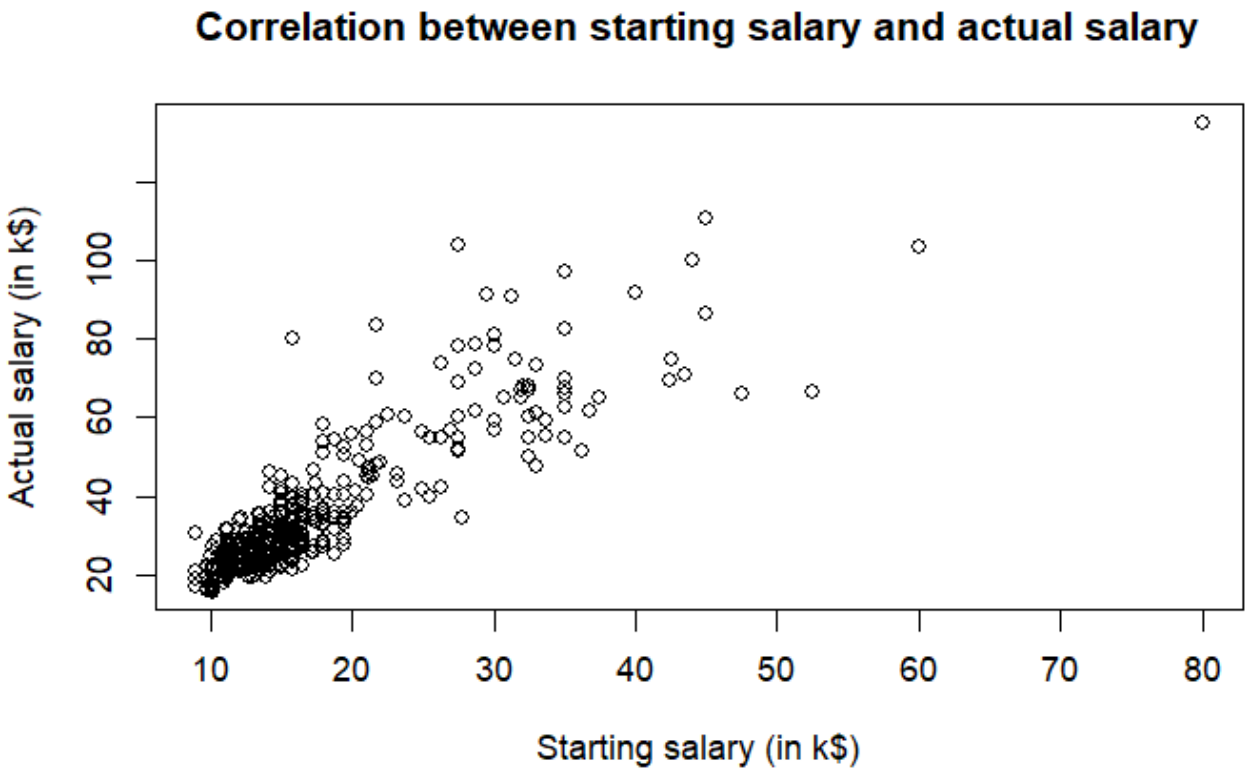


$\hookrightarrow$ get a linear correlation by transforming one or both variables (e.g. logarithm, exponential or square functions)

2.1 Scatter plot (cont.)

In order to illustrate the relationship between two quantitative variables with the data set *employees*, we are interested in the relationship between current salary (variable *salary*) and starting salary (variable *start_sal*).

The dependent variable is *salary* and the independent variable is *start_sal*.



2.2 Covariance

Let X and Y be two quantitative variables.

Definition (Covariance between X et Y $Cov(X, Y)$)

Measures the sens of the correlation between X and Y.

$$Cov(X, Y) = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

Interpretation :

- If $Cov(X, Y) = 0$, X and Y are not linearly correlated
- If $Cov(X, Y) > 0$, **positive** linear correlation between X and Y as the value of X increases, the value of Y tends to increase
- If $Cov(X, Y) < 0$, **negative** linear correlation between X and Y as the value of X increases, the value of Y tends to decrease

unit: product of units of X and Y (not to be specified).

2.2 Covariance (cont.)

König-Huygens Formula for covariance

$$Cov(X, Y) = \frac{1}{n}\sum_{i=1}^{n} x_i y_i - \bar{x}\bar{y}$$

mean of products - product of the means

Starting salary and salary in *employees* data set

Let X be the variable *start_sal* and Y be the variable *salary*. We give the following results:

$$\sum_{i=1}^{n} x_i = 8,045.375 \quad \sum_{i=1}^{n} y_i = 16,279.92 \quad \sum_{i=1}^{n} x_i y_i = 332,856.2$$

Calculate the covariance between X and Y and interpret its value.

2.2 Covariance (cont.)

Properties

- $Cov(X, Y) = Cov(Y, X)$
- $Cov(X, X) = Var(X)$
- Affine change of variable
  for $a, b, c, d \in \mathbb{R}$, $Cov(a\ X + b, c\ Y + d) = ac\ Cov(X, Y)$

2.3 Linear correlation coefficient

Definition (Linear correlation coefficient $r(X, Y)$)

Measures sense and intensity of a **linear** correlation between two quantitative variables X et Y. $-1 \leqslant r(X, Y) \leqslant 1$

$$r(X, Y) = \frac{Cov(X, Y)}{\sigma_X \sigma_Y}$$

with:

- $Cov(X, Y) = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$ the covariance between X and Y
- σ_X and σ_Y the standard deviation of X and Y, $\sigma_X = \sqrt{Var(X)}$

2.3 Linear correlation coefficient (cont.)

Interpretation

1. *sign* = sense
   (if positive, positive correlation; if negative, negative correlation)
2. *absolute value* = strength
   (if close to 1, strong linear correlation)

Detailed interpretation

- $r(X, Y) = 1$: perfect positive linear correlation between X and Y: $Y = aX + b$ with $a > 0$ (scatter plot 1)
- $r(X, Y) = -1$: perfect negative linear correlation between X and Y: $Y = aX + b$ with $a < 0$ (scatter plot 2)
- $r(X, Y)$ close to 1 (-1 resp.): strong positive (negative resp.) linear correlation between X and Y (scatter plots 4 et 5)
- $r(X, Y)$ close to 0: no linear correlation between X and Y (scatter plot 3) but another type could be possible (scatter plot 6)

2.3 Linear correlation coefficient: Example

Starting salary and salary

We give the following results:

$$\sum_{i=1}^{n} x_i^2 = 166,136.2 \quad \sum_{i=1}^{n} y_i^2 = 698,245.9$$

Calculate, the linear correlation coefficient $r(X, Y)$.

2.4 Pearson correlation test

r is the Pearson correlation of the **sample**. It is an estimate of ρ, the Pearson correlation of the **population**, the true correlation coefficient which is unknown.

Test hypotheses:
H_0 : $\rho = 0$ (no linear correlation between X and Y)
H_1 : $\rho \neq 0$ (linear correlation between X and Y)

Condition of validity: X and Y follow normal distributions (or n is large).

Test statistic: $T = \sqrt{n-2}\dfrac{r}{\sqrt{1-r^2}}$
where r is the linear correlation coefficient between X and Y in the sample.

Under H_0, $T \sim \mathcal{T}(n-2)$
(Student's t distribution with $n-2$ degrees of freedom)

2.4 Pearson correlation test (cont.)

Test hypotheses:
H_0 : $\rho = 0$ (no linear correlation between X and Y)
H_1 : $\rho \neq 0$ (linear correlation between X and Y)

Decision:

- **Critical value approach**
  reject H_0 if $|t_{obs}| > t_{1-\frac{\alpha}{2}}$
  where $t_{1-\frac{\alpha}{2}}$ is the critical value of a Student's t distribution with $n-2$ df.
- **p-value approach**
  reject H_0 if p-value $\leqslant \alpha$

If H_0 is rejected, we can conclude that there is a significant linear correlation between X and Y.

2.4 Pearson correlation test (cont.)

Test hypotheses:
H_0 : $\rho = 0$ (no linear correlation between X and Y)
H_1 : $\rho \neq 0$ (linear correlation between X and Y)

```
> cor.test(start_sal,salary)

        Pearson's product-moment correlation

data:  start_sal and salary
t = 40.259, df = 471, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.8581979 0.8990569
sample estimates:
      cor
0.8802488
```

Plan

1. Hypothesis tests

2. Relationship between two quantitative variables

3. Relationship between two categorical variables
3.1 Contingency table
3.2 Chi-Square Test of Independence
3.3 Conditional percents

4. Relationship between a quantitative variable and a categorical variable

3.1 Relationship between two categorical variables: Contingency table

Data concerning two categorical variables X and Y can be displayed in a two-way *Contingency table*.

In rows: categories of the variable X,
In columns: categories of the variable Y.

This table displays *frequencies* observed in the sample, denoted n_{ij} = number of observations of category i of X and category j of Y (i for rows and j for columns).

The table is completed with row totals and column totals. We denote $n_{i.}$ the row total of category i of X and $n_{.j}$ the column total of category j of Y.

3.1 Contingency table: Example

In order to illustrate the relationship between two categorical variables with the data set *employees*, we are interested in the relationship between gender (variable *gender*) and professional status (variable *pro_status*).

Contingency table between *gender* et *pro_status*

	pro_status			
gender	Worker	Security agent	Manager	Sum
Female	206	0	10	216
Male	156	27	74	257
Sum	362	27	84	473

3.2 Chi-Square Test of Independence

We define *expected frequency* for (i, j) as $e_{ij} = \frac{n_{i.} n_{.j}}{n}$.

It is the number of expected observations of category i of X and category j of Y under the assumption of independence between X and Y (= if there is not a relationship between X and Y).

```
> round(chi2$expected,2)
        pro_status
gender   Worker Security agent Manager
  Female 165.31           12.33   38.36
  Male   196.69           14.67   45.64
```

3.2 Chi-Square Test of Independence

There is *independence* between X and Y when observed frequencies n_{ij} are equal to expected frequencies e_{ij} for each i and j.

The *Chi-Square* χ^2 measures the degrees of independence between X and Y and is defined by:

$$\chi^2 = \sum_{i=1}^{I} \sum_{j=1}^{J} \frac{(n_{ij} - e_{ij})^2}{e_{ij}}$$

Interpretation : $\chi^2 = 0 \iff$ independence between X and Y.

3.2 Chi-Square Test of Independence

Test Hypotheses:
H_0: Independence between X and Y
H_1: No independence between X and Y

Condition of validity: $e_{ij} > 5$, otherwise we use the Fisher's exact test.

Test Statistic: $\chi^2 = \sum_{i=1}^{I} \sum_{j=1}^{J} \frac{(n_{ij} - e_{ij})^2}{e_{ij}}$

Under H_0, $\chi^2 \sim \chi^2((I-1)(J-1))$
(χ^2 distribution with $(I-1)(J-1)$ degrees of freedom)

Decision:

- **Critical value approach**
  reject H_0 if $\chi^2_{obs} > q_{1-\alpha}$
  where $q_{1-\alpha}$ is the critical value of a χ^2 distribution with $(I-1)(J-1)$ df.
- **p-value approach**
  reject H_0 if p-value $\leqslant \alpha$

If H_0 is rejected, we can conclude that there is a significant relationship between X and Y.

3.2 Chi-Square Test of Independence

Test Hypotheses:
H_0: Independence between X and Y
H_1: No independence between X and Y

```
> chi2=chisq.test(gender,pro_status)
> chi2

        Pearson's Chi-squared test

data:  gender and pro_status
X-squared = 79.713, df = 2, p-value < 2.2e-16
```

3.2 Chi-Square Test of Independence

If the χ^2 test is significant, we analyze the highest contributions to χ^2, denoted c_{ij}.

contribution to χ^2 of (i,j) : $c_{ij} = \frac{(n_{ij}-e_{ij})^2}{e_{ij}}$.

If

- $n_{ij} > e_{ij}$, the cell (i,j) is **over-represented** in the sample compared to independence,
- $n_{ij} < e_{ij}$, the cell (i,j) is **under-represented** in the sample compared to independence.

Note that χ^2 is the sum for all i and all j of contributions c_{ij}.

```
> round(chi2$residuals^2,2)
        pro_status
gender   Worker Security agent Manager
  Female  10.02          12.33   20.97
  Male     8.42          10.36   17.62
```

3.3 Conditional percents

From the contingency table, we can calculate conditional percents that are useful to graphically represent the two variables.

Row-Profile or *distributions of professional status conditioning to gender*

	pro_status		
gender	Worker	Security agent	Manager
Female	0.954	0.000	0.046
Male	0.607	0.105	0.288

For each row i, we divide n_{ij} by the row total $n_{i.}$ (the contingency table is read by row).

Column-Profile or *distributions of gender conditioning to prof. status*

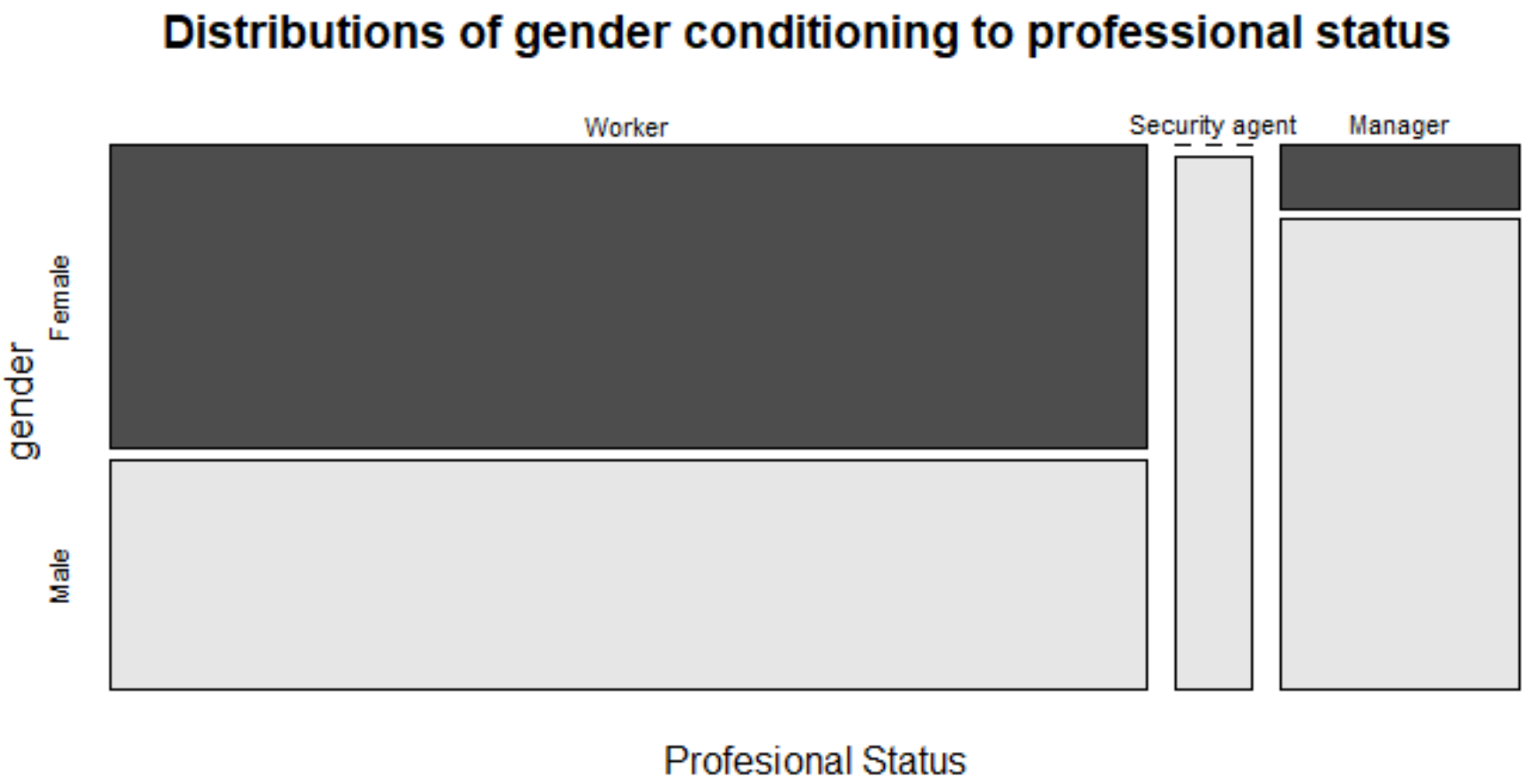
	pro_status		
gender	Worker	Security agent	Manager
Female	0.569	0.000	0.119
Male	0.431	1.000	0.881

For each column j, we divide n_{ij} by the column total $n_{.j}$ (the contingency table is read by column).

3.3 Conditional percents: graphs

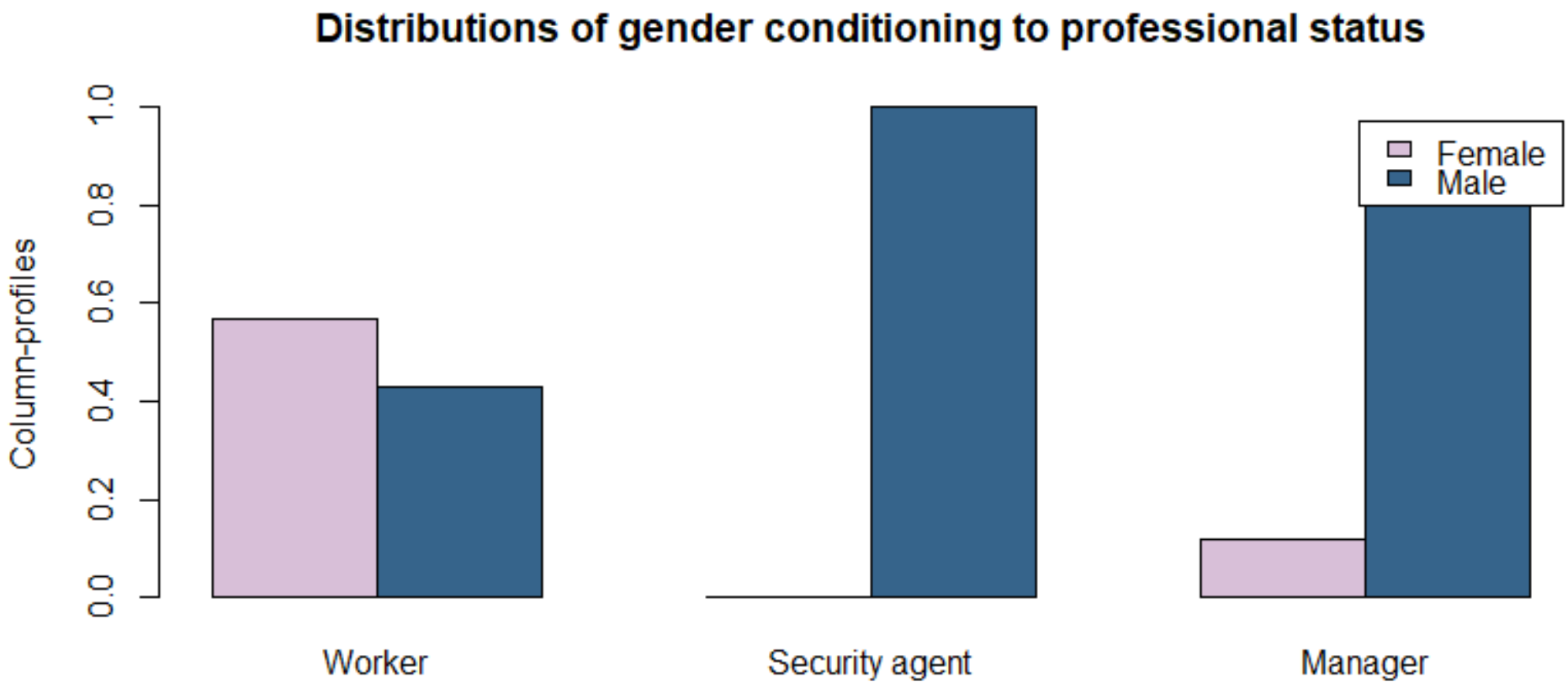
Two types of graphs may represent the conditional percents: *mosaic plots* and *multiple bar charts (side-by-side or stacked)*.

Distributions of gender conditioning to prof. status: Mosaic Plot



3.3 Conditional percents: graphs

Distributions of gender conditioning to prof. status:
Side-by-side multiple bar charts



3.3 Conditional percents: graphs

Distributions of gender conditioning to prof. status:
Stacked multiple bar charts

